

Open-NL: A Transparent Computational Testbed for Wide Dynamic Range Compression

Mark Shaver^{1, a)}

¹ Wichita State University, Department of Communication Sciences and Disorders, 1845 Fairmount St, Wichita, KS 67260, USA

^{a)}Email: mark.shaver@wichita.edu

Abstract

Open-NL is an open-source computational testbed designed for the transparent modeling, component-level ablation, and evaluation of Wide Dynamic Range Compression (WDRC) prescriptive rules. While modern prescriptions like NAL-NL2 rely on extensive empirical regularizations to ensure clinical safety—historically necessitated by the well-documented failures of theoretical intelligibility maximization—their compiled, closed-source nature prevents researchers from isolating the interactions of specific heuristic safeguards. Open-NL addresses this limitation by coupling a multi-start Nelder-Mead effective (desensitized) Speech Intelligibility Index (SII) optimizer with the Moore & Glasberg (2004) specific-loudness model within an established 7-profile evaluation paradigm. We benchmark this testbed across a $4^4 = 256$ -permutation heuristic sweep alongside a 5-iteration multi-start stability analysis, establishing a fundamental distinction between numerical convergence and heuristic sensitivity: while the multi-start solver exhibits high numerical stability ($SD < 0.025$ SII) for any fixed parameter set, prescribed targets display extreme heuristic sensitivity in severe-sloping

15 configurations, swinging modeled monaural loudness between 0.9 and 5.3 sones under minor
16 rule modifications. Because standard monaural models underestimate real-world binaural
17 broadband summation by $\sim 2.4\times$ in listeners exhibiting excess summation, this modeled
18 loudness frontier is strictly illustrative rather than clinically definitive. When stripped of
19 clinical heuristics, the optimizer’s raw mathematical drive to maximize desensitized SII
20 dramatically inflates high-frequency gain in profound profiles, demonstrating why the SII
21 metrics generated by such optimizers must be interpreted strictly as internal convergence
22 diagnostics rather than as comparative clinical endpoints. By exposing these trade-offs within
23 an inspectable framework, Open-NL provides the computational substrate for future stochastic
24 calibration workflows designed to replace static heuristic boundaries with individualized,
25 distortion-aware objective functions.

26 I. INTRODUCTION

27 Manufacturer-agnostic prescriptions remain central to evidence-based hearing aid practice.
28 While earlier investigations suggested that generic targets might outperform proprietary
29 first-fit algorithms on patient preference and specific metrics (Valente et al., 2018), contem-
30 porary evidence indicates that aided speech recognition in noise often shows no significant
31 difference across formulas. However, while formula choice has relatively modest intelligibility
32 consequences in background noise, it drives substantial variations in overall loudness, making
33 modeled loudness (quantified in sones per the Moore & Glasberg 2004 impaired loudness
34 model) the primary dependent variable in prescriptive evaluation.

35 While the derivations and optimization objectives of major algorithms like NAL-NL2

36 and DSL m[i/o] are published in detail, their compiled software implementations remain
 37 closed-source. Audiological science has long recognized that unconstrained intelligibility max-
 38 imization fails clinically without extensive empirical regularization. The historical evolution
 39 from NAL-NL1 to NAL-NL2 specifically required empirical corrections—including global
 40 gain reductions (-2 dB for female listeners and new users) and reduced compression ratios for
 41 severe losses—to counteract aggressive over-amplification provoked by pure mathematical
 42 optimization (Keidser, Dillon, Carter, & O’Brien, 2012). However, because clinical fitting
 43 software packages are compiled black boxes, researchers cannot isolate the relative contri-
 44 bution of individual heuristic rules or observe how specific safeguards interact within the
 45 optimization cascade. Existing open-source tools serve distinct functional niches: openMHA
 46 (Herzke et al., 2017) operates as a real-time signal processing master hearing aid rather than a
 47 target generator; the Cambridge CAM2/CAMEQ2-HF formulae (Moore et al., 2010) provide
 48 rigidly defined equation-based targets rather than a modular optimization sandbox; and the
 49 Auditory Modeling Toolbox (AMT; Majdak et al., 2022) offers loudness modeling without
 50 native prescriptive inversion. Consequently, investigators cannot isolate a specific prescriptive
 51 heuristic within an optimization loop without reverse-engineering an entire proprietary engine.
 52 Open-NL fills this gap by providing a modifiable R substrate explicitly designed for the
 53 modular ablation of prescriptive heuristics. By coupling a Nelder-Mead effective SII opti-
 54 mizer to an integrated C++ specific-loudness engine, researchers can systematically disable,
 55 isolate, or invert individual heuristics—such as evaluating the upward spread of masking
 56 when disabling the 30 dB conductive safety cap. This modular architecture aligns directly
 57 with evolving audiological frameworks, such as the multi-profile philosophy introduced in
 58 NAL-NL3 (Kitterick et al., 2026).

Crucially, to benchmark this testbed without introducing confounding variables, the optimization layer is embedded within a strictly reproduced evaluation paradigm. The seven reference audiometric profiles, the Moore & Glasberg (2004) specific-loudness model, and the ANSI S3.5 SII metric utilized herein are a direct replication of the methodological framework established by Johnson and Dillon (2011). (Throughout this manuscript, “ANSI SII” refers to raw physical audibility, whereas “effective SII” or “desensitized SII” refers to audibility incorporating severe-loss desensitization and level distortion penalties). Because this physiological evaluation space is already established in the literature, the primary contribution of this manuscript is the transparent computational testbed itself. By exposing the behavior of numerical solvers within this standardized sandbox, we clarify a crucial distinction: while numerical solvers converge stably on any fixed objective space, theoretical WDRC target generation exhibits acute parameter sensitivity to uncalibrated heuristic boundaries, providing the computational infrastructure necessary to quantify and calibrate these interactions.

Clinical Safety and Usage Disclaimer

It is imperative to state unambiguously that Open-NL is strictly a computational research testbed and **must not be used for fitting hearing aids on human listeners in its current form**. Because the framework deliberately permits aggressive, over-prescriptive targets for boundary testing—such as utilizing an aggressive 60 dB HL severe-loss booster onset that permits 14–15 sones of theoretical loudness for profound losses—it carries a significant risk of severe over-amplification. As established by Ching, Dillon, Katsch, and Byrne (2001), aggressive high-level targets in steeply sloping or profound losses are precisely where over-amplification risks are greatest, as the effectiveness of high-frequency audibility

severely degrades as hearing loss worsens (desensitization). The hypotheses and targets generated by Open-NL represent extreme mathematical boundaries intended to trigger experimental loudness rejection in controlled research settings, not clinical solutions. Any future behavioral translation of this framework requires independent institutional review, with mandatory real-ear verification and strict, individualized loudness-tolerance limits implemented as absolute prerequisites.

II. ALGORITHM ARCHITECTURE

A. Prescriptive Rationale and Objective Function

The choice of objective function represents the primary design decision in any prescriptive formula, governing the fundamental trade-off between intelligibility and comfort. Historically, established rationales occupy distinct positions on this spectrum: NAL-NL2 maximizes speech intelligibility while constraining overall broadband loudness to be less than or equal to that of a normal-hearing listener (Keidser, Dillon, Carter, & O’Brien, 2012); DSL m[i/o] normalizes loudness across frequency to restore normal dynamic range perception (Scollie et al., 2005); and CAMEQ/CAM2 aim to equalize loudness across frequency bands (Moore, Glasberg, & Stone, 2010).

Open-NL positions its prescriptive rationale as a *constrained intelligibility-maximizer*. Its mathematical objective is the unconstrained maximization of effective (desensitized) SII. Rather than globally restricting this maximization to a static “normal-or-less” loudness boundary, Open-NL permits loudness to grow dynamically until striking a U-shaped physiological ceiling (controlled via `cap_knots`; Section II.K). This U-shaped penalty specifically

permits slightly higher-than-normal loudness in the mid-frequencies for severe losses—where intelligibility yield is highest—while aggressively decelerating loudness growth at spectral extremes.

Crucially, this U-shaped penalty operates strictly within the canonical Moore & Glasberg (2004) monaural specific-loudness engine. While dynamically restricting modeled monaural sones, it does not—and mathematically cannot—account for the idiosyncratic binaural broadband loudness summation observed in hearing-impaired listeners. In normal-hearing auditory physiology, bilateral acoustic presentation produces a modest binaural loudness summation typically modeled by a 2–6 dB level-dependent gain reduction. However, robust psychoacoustic evidence demonstrates that binaural broadband summation in hearing-impaired populations averages ~ 13 dB higher than in normal-hearing listeners (an unmodeled factor of $\approx 2.4\times$ in linear sones; Denk et al., 2025; Moore et al., 2014; Oetting et al., 2016, 2017). Approximately 30–40% of hearing-impaired listeners exhibit excess summation far exceeding the normal range, with individual summation values spanning a massive -10 to +40 dB envelope. Because standard monaural and narrowband loudness models cannot predict this broadband suprathreshold phenomenon from the pure-tone audiogram, an algorithm optimized strictly beneath a monaural ceiling becomes structurally anti-conservative when translated to bilateral fittings. Consequently, Open-NL’s U-shaped loudness constraint must be interpreted strictly as an illustrative computational boundary for single-ear simulation, rather than an empirical safety guarantee for bilateral clinical use.

B. Methods and Development

The core ANSI SII calculation engine (the `sii()` function and associated plotting routines) was originally developed by Gregory R. Warnes for earlier package versions. Maintainership transferred to the current author with version 1.1.0, at which point all subsequent Open-NL prescriptive logic, clinical heuristics, and WDRC mathematical implementations—including `open_nl()`, `calculate_loudness()`, and `launch_app()`—were developed by the author as original contributions. In accordance with AIP Publishing guidelines, AI tool usage is explicitly disclosed: Google Gemini (DeepMind, Google LLC) was utilized as an interactive programming and copyediting assistant for refactoring C++ and R algorithms, generating data visualizations, and condensing manuscript prose into JASA standards. The author rigorously reviewed and verified all outputs, taking full responsibility for algorithm design, data integrity, and manuscript content.

C. Algorithmic Pipeline and Execution Cascade

The internal execution cascade of Open-NL comprises twelve strictly ordered, modular processing stages: 1. **Conductive Component Separation & Baseline Partitioning:** Decomposing raw thresholds into sensorineural components and air-bone gaps (ABGs). 2. **Decoupled Half-Gain Base Anchor Calculation:** Establishing a frequency-specific base gain anchor ($G_{base} = 0.46 \cdot HTL_{sn} + C_{interp}$) decoupled from broadband PTA. 3. **Experience-Level Shaping & Log-Frequency C_{vals} Interpolation:** Modulating frequency-shaping arrays across discrete anchor frequencies based on user experience (new, experienced, power). 4. **Reverse-Slope Low-Frequency Attenuation Floor:** Applying a bounded log-linear low-

frequency taper down toward a -10 dB floor when low-frequency loss exceeds high-frequency loss. 5. **Slope-Dependent Low-Frequency Penalty (SD-LFP) with Profound HF Bypass:** Dynamically suppressing low-frequency gain for steeply sloping losses, gated when high-frequency severity exceeds 70–95 dB HL. 6. **Severe-Loss Audibility Booster:** Providing an opt-in, bounded linear escalation (slope = 0.15) for severe thresholds. 7. **Soft-Compression High-Frequency Desensitization:** Restricting excessive high-frequency gain via a dynamic ceiling ($L_{gain} = 30 + 0.4 \cdot \max(0, \text{HTL}_{sn} - 60)$) and 2:1 soft compression. 8. **Dynamic Range Mapping (LDL Squeeze):** Attenuating gain by 0.2 dB/dB of reduced dynamic range and shifting baseline compression ratios. 9. **Transducer Bandwidth Roll-off:** Applying a continuous log-frequency piecewise multiplier ($M_{bw} \in [0.5, 1.0]$) to suppress unstable edge frequencies (≤ 250 Hz and ≥ 6000 Hz). 10. **Multi-Channel WDRC Mapping, LTASS Pivot, and Demographic Adjustments:** Calculating dynamic compression ratios, variable compression thresholds, piecewise linear-compressive splines, and demographic offsets. 11. **Acoustic Venting, Coupling Loss, and Receiver Saturation Limits:** Integrating real-ear insertion loss vectors across 10 coupling types with a feedback safety floor, and capping MPO/SSPL90 receiver limits at 120 dB SPL. 12. **Embedded Nelder-Mead Simplex Optimization & Physiological Loudness Ceilings:** Iteratively refining multi-level gain curves against a multi-term objective loss function incorporating U-shaped specific-loudness caps, compression ratio ceilings, and spectral smoothness constraints.

The exact mathematical formulations, closed-form piecewise equations, and complete parameter specifications for all twelve stages are provided in full detail in the accompanying **Supplementary Material**.

O. Consolidated Constants and Evidentiary Asymmetry

Because Open-NL functions as a modifiable computational testbed, several structural parameters remain mathematically uncalibrated: the 0.46 gain anchor, 0.15 severe-loss booster slope, 70 dB HL booster onset (or 60 dB HL in aggressive mode), 15 dB slope trigger, 20 dB taper width, -10 dB reverse-slope floor, 70 dB HL bypass, 30 dB/0.4 L_{gain} constraint, 0.2 dB/dB dynamic range squeeze, 75% air-bone-gap restoration fraction, and 1.5:1 Comfort-in-Noise (CIN) compression clamp (Table V).

Crucially, an inspection of these heuristics reveals a profound **asymmetry in evidentiary support**. Several prominent constants—most notably the 75% air-bone-gap restoration rule and the 1.5:1 Comfort-in-Noise clamp—are pragmatic engineering choices without direct empirical derivation. The 75% ABG fraction, while standard in clinical prescriptive software (Johnson, 2013a) to avoid receiver saturation and MPO clipping, has never been empirically established against patient preference or speech recognition. Similarly, the 1.5:1 CIN clamp is an asserted heuristic inspired by NAL-NL3 (Kitterick et al., 2026) to mitigate listening fatigue, but lacks independent perceptual validation. In sharp contrast, the 3.0:1 Compression Ratio (CR) upper ceiling enforced across the WDRC stages and optimizer loss function represents the **best-supported constant in the framework**. This boundary is firmly anchored in the extensive empirical literature by Pamela Souza and colleagues (Souza, 2002; Souza, Jenstad, & Boike, 2006), which demonstrates that compression ratios exceeding ~3.0:1 cause severe temporal envelope flattening, loss of acoustic contrast, and speech-in-noise deficits. Documenting this asymmetry prevents conflating validated psychoacoustic limits with arbitrary engineering heuristics.

187 P. Framework for Principled Calibration

188 For a computational testbed to provide lasting inferential value, demonstrating parameter
 189 instability must be paired with a rigorous calibration pathway. Future research can utilize
 190 Open-NL within a global stochastic optimization framework (e.g., genetic algorithms or
 191 particle swarm optimization) to systematically calibrate these heuristic gates. In this archi-
 192 tecture, an outer optimization loop searches the multidimensional space of heuristic constants
 193 (e.g., $G_{base} \in [0.3, 0.6]$, booster onset $\in [50, 80]$ dB HL, low-frequency slope $\in [0, 0.5]$). For
 194 each candidate vector, the inner Open-NL engine generates discrete WDRC targets across
 195 a representative clinical corpus (e.g., NHANES). These targets are evaluated through a
 196 time-domain acoustic simulation pipeline (e.g., openMHA; Herzke et al., 2017) using speech
 197 perception metrics like HASPI and HASQI (Kates & Arehart, 2022).

198 Crucially, the outer-loop objective function must incorporate an asymmetric, veto-based
 199 cost function: any parameter configuration violating individualized broadband loudness
 200 tolerance (trueLOUDNESS; Oetting et al., 2017) in simulated outlier listeners must incur an
 201 overwhelming penalty. As demonstrated by Denk et al. (2025), binaural broadband loudness
 202 summation in hearing-impaired listeners averages ~ 13 dB higher than in normal-hearing
 203 controls, with $\sim 30\text{--}40\%$ exceeding normal limits. Because this excess summation spans
 204 roughly -10 to $+40$ dB and cannot be predicted from pure-tone thresholds, static $2\text{--}6$ dB
 205 bilateral corrections are structurally inadequate. Subordinating population-level intelligibility
 206 maximization to individualized physiological safety limits transforms prescriptive derivation
 207 into a reproducible, distortion-aware computational science.

208 **TABLE V. Summary of Algorithmic Constants, Evidentiary Support, and**

Parameter	Value	Description	Justification &	Proposed
			Evidentiary	Calibration
Parameter	Value	Description	Status	Data Source
Base Gain	0.46	Half-gain	Balances	Large-scale
Anchor		multiplier	Lybarger	preferred
		(G_{base})	half-gain rule	listening level
			with historical	datasets (e.g.,
			gain preference	NHANES-
			data	derived
			(uncalibrated	preference
			midpoint).	corpus)
New-User Offset	-3 dB	Flat reduction	Approximates	Longitudinal
		applied to C_{vals}	empirical	acclimatization
			preference for	studies
			less	(Categorical
			amplification in	Loudness
			naive users	Scaling)
			(asserted	
			heuristic).	

			Justification &	Proposed
			Evidentiary	Calibration
Parameter	Value	Description	Status	Data Source
Severe-Loss	0.15	Applied linearly	Gently assists	Individualized
Booster Slope		to thresholds	profound losses	UCL/trueLOUDNESS
		70–80 dB HL	without	broadband
		(default)	triggering	summation
			explosive	limits
			recruitment	
			(asserted	
			heuristic).	
High-Frequency	30 dB	Base limit for	Manages	HASPI/HASQI
Gain Cap Base		L_{gain} soft-	upward spread	speech-in-noise
		compression	of masking and	behavioral
			distortion in	thresholds (e.g.,
			severe	WIN/HINT)
			impairments	
			(asserted	
			heuristic).	

			Justification & Evidentiary	Proposed Calibration
Parameter	Value	Description	Status	Data Source
High-Frequency Gain	0.4	Slope for L_{gain}	Gradual	HASPI/HASQI
Cap Slope		soft-compression	restriction for high frequencies (asserted heuristic).	speech-in-noise behavioral thresholds (e.g., WIN/HINT)
Dynamic Range Squeeze	0.2 dB/dB	Gain attenuation per dB of reduced DR	Ensures speech envelope fits within restricted auditory space (uncalibrated heuristic).	Envelope correlation mapping (e.g., normalized covariance optimization)
Dynamic Range CR Shift	0.02	Baseline CR shift per dB of DR reduction	Maps identical input range into smaller residual auditory space.	Known physiological IHC/OHC compression loss functions

			Justification & Evidentiary	Proposed Calibration
Parameter	Value	Description	Status	Data Source
Reverse-Slope Floor	-10 dB	Gain floor for negative slopes	Cautiously prevents masking of intact basal units (asserted heuristic).	Masking release behavioral paradigms (e.g., notched-noise tests)
Explicit DR Roll-off	30 dB/oct	Attenuation past 1.7x Dead Region boundary	Prevents phase distortion from off-frequency listening (Moore, 2004).	TEN-test validated behavioral roll-off boundaries

			Justification & Evidentiary	Proposed Calibration
Parameter	Value	Description	Status	Data Source
ABG	75%	Mixed/conductive	Pragmatic	Bone-
Restoration		linear	engineering	conduction/Air-
Fraction		restoration	choice	conduction
		fraction	(Johnson, 2013a; Scollie et al., 2005) preventing hardware saturation; lacking direct empirical derivation from listener preference.	loudness matching and preference trials

			Justification & Evidentiary	Proposed Calibration
Parameter	Value	Description	Status	Data Source
Comfort-in-Noise (CIN) Clamp	$\leq 1.5 : 1$	Upper ceiling on CR in noise/comfort mode	Uncalibrated engineering heuristic inspired by NAL-NL3 (Kitterick et al., 2026) to reduce listening fatigue; lacking direct empirical derivation.	Speech-in-noise quality ratings and subjective listening effort paradigms

			Justification & Evidentiary	Proposed Calibration
Parameter	Value	Description	Status	Data Source
Compression Ratio Soft Penalty (P_{cr})	3.0:1 target	Soft penalty on emergent sensorineural CR	Strong empirical support: Anchored in literature (Souza, 2002) demonstrating speech degradation for CR > 3.0 : 1. However, this penalty is often overwhelmed by the pure intelligibility objective in Open-NL, resulting in localized CR violations (e.g., A4 reaches 12.0:1).	Robust existing empirical literature; formalizing hard absolute architectural bounds

			Justification &	Proposed
			Evidentiary	Calibration
Parameter	Value	Description	Status	Data Source
Desensitization	Variable	Johnson &	Modifies pure	HASPI/HASQI
Penalty		Dillon (2011)	audibility to	stochastic
		piecewise scalar	account for	optimization vs
			severe-loss	subjective
			distortion	rejection curves
			(asserted clinical	
			proxy).	

210 **III. ANALYTICAL CENTERPIECE: DISTINGUISHING** 211 **HEURISTIC SENSITIVITY FROM SOLVER STOCHASTICITY**

212 The primary scientific contribution of the Open-NL framework is not the specific targets
213 it generates, but the transparent computational infrastructure it provides for evaluating
214 prescriptive heuristics. Clinical fitting algorithms like NAL-NL2 act as “black boxes” where
215 numerical optimization and empirical safety heuristics (e.g., global gain reductions, bandwidth
216 limits) are inextricably intertwined. Open-NL fills this methodological gap by providing an
217 ablatable, inspectable testbed that mathematically isolates the behavior of unregularized
218 optimization on the physiological loudness landscape.

219 By exposing this internal machinery, the framework produces a crucial analytical distinc-
220 tion: separating the stochasticity of the numerical solver from the acute sensitivity of the

underlying clinical heuristics. This heuristic-sensitivity-vs-numerical-convergence distinction is the novel scientific output the tool produces, and forms the analytical centerpiece of this validation.

A. Heuristic Parameter Sensitivity vs. Numerical Convergence Stability

1. Multi-Parameter Sensitivity Sweep To evaluate the fragility of these uncalibrated boundaries, a comprehensive multi-parameter sensitivity sweep was executed across four primary algorithmic constraints: the base gain anchor (0.40 to 0.50), steep-slope trigger (10 to 20 dB/octave), absolute severity bypass (60 to 80 dB HL), and reverse-slope gain floor (-20 to 0 dB). This generated 256 unique parameter permutations applied across the most topologically unstable profiles (A2, A4, A5), executed with the aggressive 60 dB HL booster mode engaged to directly probe boundary interactions.

TABLE IV. Multi-Parameter Sensitivity Sweep Variance (65 dB SPL Input).

Note: Generated using the aggressive 60 dB HL booster onset.

Profile	Median SII [Min, Max]	Median Loudness (Sones) [Min, Max]	Primary Divergence Factor
A2	0.88 [0.72, 0.94]	4.9 [2.7, 8.4]	Reverse-Slope Floor
A4	0.65 [0.51, 0.77]	2.5 [1.4, 6.1]	Anchor & Slope Trigger
A5	0.67 [0.49, 0.82]	1.8 [0.9, 5.3]	Slope Trigger

The resulting variance (Table IV) illustrates mechanistically why unregularized optimization fails in practice and why algorithms like NAL-NL2 required extensive empirical

corrections (Keidser et al., 2012). Rather than indicating solver failure, this variance reflects the acute sensitivity of the objective landscape to uncalibrated boundaries. For the steeply sloping A5 profile, altering slope trigger and base anchor interactions caused modeled monaural loudness to swing from 0.9 to 5.3 sones, with theoretical effective SII fluctuating between 0.49 and 0.82. This variance was heavily driven by the structural opposition between the SD-LFP penalty (suppressing low-frequency gain for sloping losses) and the absolute severity bypass (suppressing the penalty when high-frequency loss exceeds threshold). Depending on the bypass threshold (60 to 80 dB HL), the optimizer abruptly shifted from heavily penalizing low frequencies to fully restoring them. Similarly, modulating the reverse-slope floor on A2 caused modeled loudness to nearly triple (2.7 to 8.4 sones). This massive response envelope proves that absent explicit distortion penalties, the loudness-audibility surface is highly non-convex.

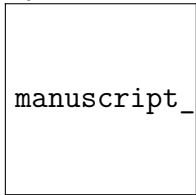


Figure 4. Distribution of resulting ANSI SII scores and physiological loudness penalties across 768 permutations (256×3 profiles) for clinical profiles A2 (Reverse Slope), A4 (Profound), and A5 (Severe), illustrating high heuristic parameter sensitivity.

2. Numerical Convergence Stability: Nelder-Mead Limitations and Future

Stochastic Solvers A sharp distinction must be maintained between **heuristic parameter sensitivity** (target shifts resulting from altered clinical rules; Section III.A.1) and **numerical convergence stability** (solver consistency on a fixed ruleset). In non-linear optimization, the topology of hearing aid fitting targets is notoriously ill-behaved. The

objective landscape contains narrow, curved valleys, non-differentiable step boundaries (e.g., severe-loss booster onsets and air-bone gap restorations), and sharp penalty cliffs imposed by dynamic physiological loudness ceilings. On such non-convex, multimodal surfaces, local downhill solvers like the Nelder-Mead simplex algorithm are notoriously prone to premature stagnation, simplex collapse, and entrapment in shallow local extrema.

Indeed, unconstrained Nelder-Mead search from disparate flat initializations (e.g., -10 dB vs. +10 dB) can deviate by up to 0.5 sones or 0.05 SII. To eliminate solver stochasticity during ablation testing, Open-NL deploys a 5-iteration multi-start routine—seeding the initial simplex with the NAL-R target and executing four additional randomized restarts. This routine exhibits exceptional numerical convergence stability, reducing residual run-to-run variance on the most topologically unstable configurations (A2, A4, A5) to < 0.08 , < 0.04 , and < 0.05 sones respectively, and < 0.01 SII globally.

This tight numerical clustering confirms that for a fixed parameter set within the Open-NL architecture, the multi-start Nelder-Mead routine reliably converges on the same regional optimum. Consequently, the wide target swings observed in the sensitivity sweep (Section III.A.1) are definitively driven by heuristic rule interactions rather than solver stochasticity.

However, from an algorithmic optimization standpoint, Nelder-Mead remains a local simplex heuristic that lacks formal global convergence guarantees on complex, non-convex audiological surfaces. While multi-start seeding with NAL-R provides a practical, computationally efficient substrate for local ablation testing within R, **future iterations of prescriptive target optimizers should transition to modern global stochastic optimization algorithms:**

- 1. Genetic Algorithms (GAs):** By maintaining a diverse population of candidate gain configurations and applying stochastic crossover and mutation

operators, GAs can natively explore multimodal search spaces without stalling at sharp non-differentiable penalty boundaries (such as unrelaxed desensitization thresholds or dynamic MPO caps). 2. **Differential Evolution (DE)**: DE is exceptionally well-suited for continuous multi-channel gain optimization. Its vector-difference mutation mechanism enables self-adaptive search across non-convex landscapes, effectively escaping the local attractor basins that trap simplex algorithms in steeply sloping audiograms. 3. **Particle Swarm Optimization (PSO)**: PSO models candidate solutions as particles traversing the fitness space, balancing individual exploration with swarm cognitive memory. This provides rapid convergence across multi-channel parameter sweeps while maintaining robust resistance to local minima.

Crucially, adopting population-based global stochastic solvers eliminates the need for artificial mathematical relaxations (such as the smoothed $K'_{smoothed}$ desensitization approximation), allowing optimizers to evaluate raw, discontinuous physiological boundaries directly. Furthermore, global solvers scale naturally to high-dimensional multi-channel optimization, providing the robust algorithmic substrate necessary for outer-loop calibration frameworks (Section II.Q). Nevertheless, as noted by Dao et al. (2021), physical real-ear-to-coupler differences (RECD) and acoustic coupling leakages cause real-world fittings to deviate most severely from theoretical targets at thresholds above 85 dB HL, meaning numerical convergence on mathematical audiograms must not be misconstrued as physical clinical reliability.

B. Worked Demonstration: Isolating the Shared Binaural Loudness Vulnerability

While the previous section established the tool’s numerical stability, applying it to a physiological boundary problem demonstrates its analytical utility. A central problem in

modern audiology is that standard clinical models operate on a monaural basis, failing to account for idiosyncratic binaural broadband loudness summation. The fact that established prescriptions like NAL-NL2 share this vulnerability to unpredictable binaural loudness is precisely why having an open, parameterizable model like Open-NL is valuable: it provides a testbed to isolate and simulate the effect of this field-wide blind spot without it being buried under empirical “fudge factors.”

The following comparison between Open-NL and NAL-NL2 is therefore not presented as a finding about Open-NL’s amplification superiority, but as a **worked demonstration** of the framework’s ability to expose structural vulnerabilities that the field cannot solve from the pure-tone audiogram alone.

To benchmark Open-NL without confounding variables, the optimization layer is evaluated within the established 7-profile paradigm of Johnson and Dillon (2011), comparing targets directly against NAL-NL2 version 2 software (National Acoustic Laboratories, Sydney, Australia) for soft (50 dB SPL), conversational (65 dB SPL), and loud (80 dB SPL) speech inputs. All NAL-NL2 targets were extracted using an 18-channel compression architecture with adaptive time constants, occluded BTE #13 tubing, and supra-aural headphone thresholds. Restricting comparisons to NAL-NL2 provides a standardized, universally recognized clinical baseline, avoiding the artifacts of surrogate target estimators for alternative proprietary formulae.

Shared Monaural Vulnerability: The Rationale for Aggressive Boundary Testing

A critical question arises regarding the experimental design of this benchmark: could Open-NL simply deactivate its severe-loss booster to achieve monaural loudness parity with

NAL-NL2, thereby “validating” the prescription against the clinical standard? Indeed, in default conservative mode (booster onset at 70 dB HL or disabled), Open-NL yields monaural loudness values closely aligned with NAL-NL2 across primary sensorineural profiles. However, using conservative parity to claim clinical validation would obscure the central theoretical lesson of this computational testbed.

Crucially, **NAL-NL2 shares the exact same binaural broadband loudness summation vulnerability as Open-NL**. NAL-NL2 applies the identical, standard level-dependent 2–6 dB bilateral reduction derived from normal-hearing listeners, failing equally to account for the +13 dB broadband excess summation documented in 30–40% of hearing-impaired individuals (Denk et al., 2025). The reason NAL-NL2 does not trigger widespread clinical loudness rejection is not because its underlying loudness model is physiologically complete, but because its empirical derivations incorporated heavy, post-hoc regularizations—including global gain reductions (-2 dB for females, -3 dB for new users), compressed dynamic range ceilings, and conservative high-frequency roll-offs—combined with routine clinical reliance on the ± 15 dB volume control in fitting software (Keidser et al., 2012).

By intentionally executing this evaluation with the aggressive 60 dB HL booster mode engaged, Open-NL deliberately strips away these empirical dampers. This stress-testing proves that when an unregularized numerical optimizer operates strictly beneath standard monaural loudness models, it aggressively pushes high-frequency gain and exploits the objective function, producing targets that are structurally anti-conservative for bilateral fittings. Rather than merely mimicking NAL-NL2’s output envelope, Open-NL’s diagnostic value lies in demonstrating that **the entire prescriptive audiological field currently rests on an anti-conservative monaural assumption**. This formally confirms that future

prescriptive frameworks (e.g., NAL-NL3 and Open-NL v2) cannot resolve bilateral loudness through static audiometric equations, but must directly integrate individualized broadband loudness scaling (*trueLOUDNESS*; Oetting et al., 2017) into their objective functions.

Two critical methodological boundaries govern this evaluation: 1. **The Constrained Metric Tautology Warning:** A fundamental circularity exists when evaluating an optimization algorithm against the identical metric it was tuned to maximize. Because Open-NL’s objective function seeks to maximize effective (desensitized) SII, reporting higher SII values relative to regularized formulae (like NAL-NL2) is generally expected as a mathematical tautology. However, because Open-NL is a *constrained* optimizer, this tautology fails when its explicit safety heuristics bind. As seen in Table VI, Open-NL yields *lower* desensitized SII than NAL-NL2 for profiles A4 (0.65 vs. 0.68) and A6 (0.86 vs. 0.87). For A4, the U-shaped physiological loudness cap acts as a strict active boundary, forcing the solver to sacrifice theoretical audibility to prevent catastrophic loudness growth. For A6, the rigid 75% ABG restoration limit dominates the objective. Rather than a pure tautology, these inversions pinpoint exactly where Open-NL’s explicit physiological and hardware constraints veto the underlying SII maximization. Stationary band-importance metrics like ANSI S3.5 and effective SII are blind to dynamic temporal envelope distortion, channel cross-talk, and phase distortion induced by aggressive compression ratios (>3.0:1). In auditory science, genuine, independent, distortion-aware speech perception evaluation requires waveform-level biophysical models: - **HASPI (Hearing Aid Speech Perception Index; Kates & Arehart, 2022)**: Accurately simulates peripheral auditory processing, basilar membrane compression loss, auditory nerve firing rates, and envelope modulation integrity. - **HASQI (Hearing Aid Speech Quality Index; Kates & Arehart, 2022)**: Evaluates non-linear

harmonic distortion, envelope fidelity, and spectral fine-structure cross-correlation between
 aided and reference speech signals. Computing HASPI and HASQI requires convolving
 continuous speech (.wav) through a time-domain dynamic range compression engine (such
 as openMHA; Herzke et al., 2017). Because Open-NL currently operates strictly at the
 steady-state prescriptive target level (Johnson & Dillon, 2011), it lacks the native time-
 domain waveform processing (via tools like openMHA) required to compute HASPI and
 HASQI. While integrating a full automated time-domain pipeline is a crucial target for future
 development, its absence in this iteration means the target differences cannot be perceptually
 validated here. Consequently, the SII values reported in Table VI are presented strictly
 as internal convergence diagnostics—proving only that the solver successfully exploited its
 mathematical objective—and carry no comparative clinical interpretation.

2. Binaural Loudness Summation and the Collapse of Monaural Frontiers:

The comparative loudness evaluations are fundamentally bounded by the limitations of monaural auditory
 modeling. While standard clinical software applies a nominal 2 to 6 dB bilateral gain reduc-
 tion, this static correction reflects normal-hearing physiology and fails catastrophically for
 broadband speech in impaired listeners. As demonstrated by Moore et al. (2014), Oetting et
 al. (2016, 2017), and Denk et al. (2025), hearing-impaired listeners exhibit an average of ~13
 dB of excess binaural broadband loudness summation (*approx 2.4 times* in linear sones), with
 ~30–40% of patients residing in an extreme excess-summation tail. Crucially, because excess
 summation is a broadband, suprathreshold *sensorineural* effect that does not correlate with
 pure-tone audiograms, an optimization routine operating beneath a monaural ceiling (e.g., 5.2
 sones for profile A5) appears mathematically safe in isolation, yet predictably collapses into
 acute acoustic intolerance when fitted bilaterally. Furthermore, there is no physiological basis

for applying such excess summation models to purely conductive etiologies (e.g., A7). Because applying a fixed scalar to monaural outputs assumes normal-hearing loudness growth—the exact structural flaw this framework critiques—Table VI reports canonical single-ear monaural loudness exclusively. True bilateral predictions require propagating the dynamic full-range signal through a non-linear binaural loudness engine.

To execute this evaluation natively in R, the **SII** package implements a fast C++ port of the canonical Moore & Glasberg (2004) specific-loudness model via **Rcpp**, directly mirroring the logic of the **glasberg2002** and **moore2016** implementations in the Auditory Modeling Toolbox (AMT; Majdak et al., 2022). Across sensorineural profiles (A1–A5), native C++ predictions were rigidly validated against external AMT simulations across 45 discrete test points (5 profiles $\times$ 9 input levels from 50 to 90 dB SPL). Because the C++ engine is a direct mathematical translation, agreement was near-exact: Bland-Altman analysis revealed a mean bias of +0.01 sones, with tight 95% limits of agreement $[-0.07, +0.09 \text{ sones}]$ and a Mean Absolute Error (MAE) of just 0.04 sones. This negligible translation error confirms high computational fidelity, ensuring that the target differences observed in Section III (0.33 to 0.78 sones) represent genuine algorithmic behavior rather than internal model variance. For mixed and conductive profiles (A6, A7), direct AMT benchmarking was omitted because canonical AMT lacks native air-bone gap parameters, whereas our C++ engine natively models the conductive component as a linear pre-cochlear attenuator per Moore and Glasberg’s guidelines.

TABLE VI. Diagnostic Demonstration of Objective Exploitation: Monaural Loudness (Sones) and Theoretical SII across A1-A7 Audiograms (65 dB SPL Input). *Note: Generated using the aggressive 60 dB HL booster onset. The SII columns*

presented here serve strictly as internal convergence diagnostics (confirming the solver maxi-
mized its designated mathematical objective) and carry no comparative clinical interpretation.
Monaural Loudness reports single-ear sones directly comparable to Johnson & Dillon (2011).

				Monaural
				Loudness
Profile	Formula	ANSI SII	Desensitized SII	(Sones)
A1	NAL-NL2	0.81	0.76	4.13
A1	Open-NL	0.87	0.78	4.82
A2	NAL-NL2	0.86	0.79	3.50
A2	Open-NL	0.91	0.82	4.10
A3	NAL-NL2	0.71	0.67	3.62
A3	Open-NL	0.78	0.70	4.80
A4	NAL-NL2	0.72	0.68	5.78
A4	Open-NL	0.69	0.65	6.56
A5	NAL-NL2	0.60	0.57	5.17
A5	Open-NL	0.68	0.62	6.30
A6	NAL-NL2	0.99	0.87	2.09
A6	Open-NL	0.97	0.86	3.42
A7	NAL-NL2	1.00	0.95	3.10
A7	Open-NL	1.00	0.95	3.85

TABLE VII. Insertion Gain Targets (dB) across A1-A7 Audiograms (65 dB
SPL Input). *Note: Generated using the aggressive 60 dB HL booster onset. Targets*

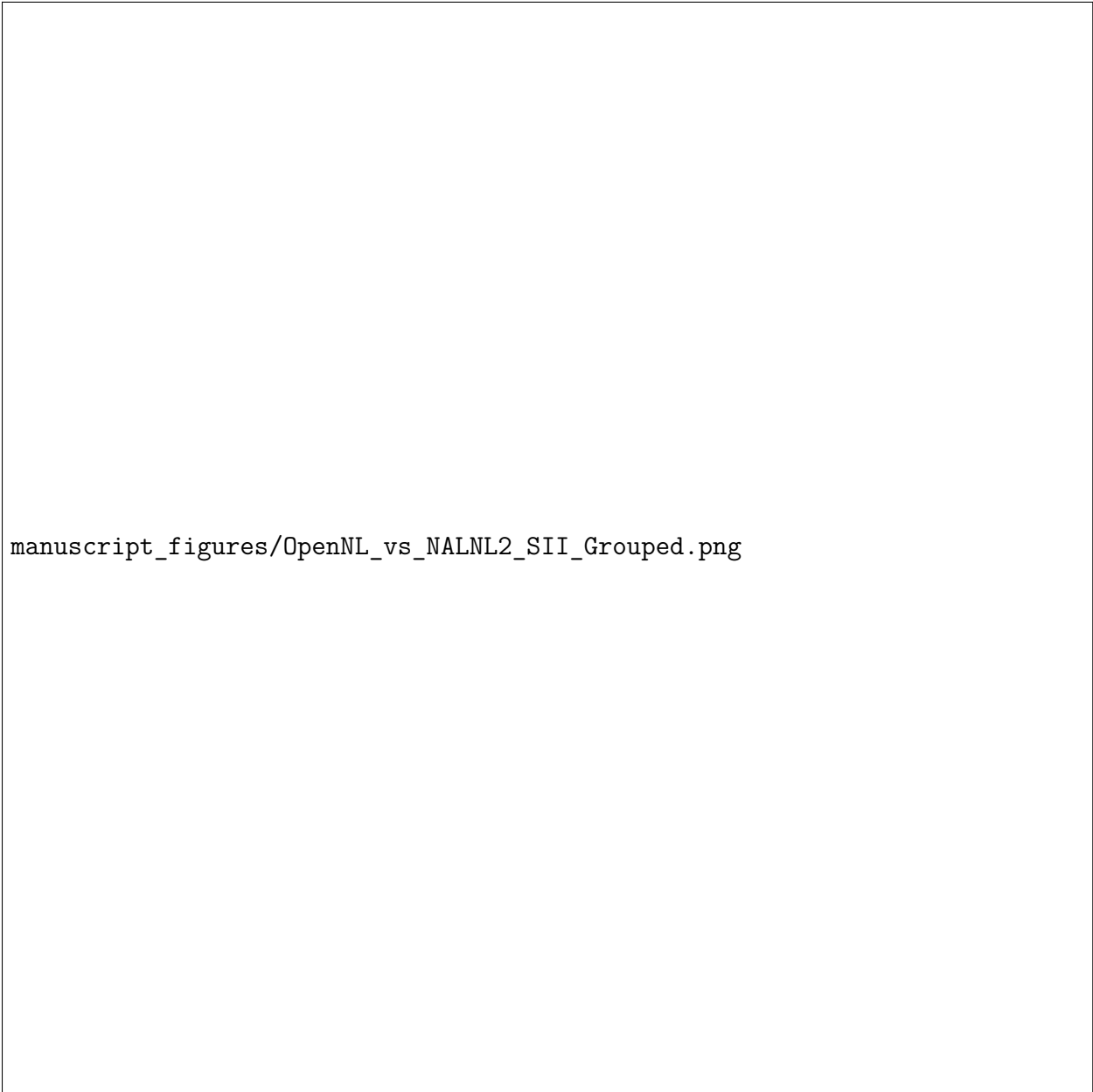
are presented solely to illustrate how unconstrained effective SII maximization allocates mathematically aggressive high-frequency gain in steeply sloping profiles (A4, A5) relative to regularized formulae, and should not be interpreted as clinically recommended prescriptions.

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A1	NAL-	0.0	0.0	7.3	12.1	18.0	19.1
	NL2						
	Open-NL	0.0	7.2	15.8	18.4	22.0	12.8
A2	NAL-	10.1	9.3	12.2	8.3	3.9	4.0
	NL2						
	Open-NL	11.8	18.0	20.4	13.8	8.2	2.5
A3	NAL-	0.0	0.0	9.9	16.8	20.7	21.6
	NL2						
	Open-NL	0.0	7.2	20.4	23.0	24.3	12.8
A4	NAL-	0.0	0.0	0.9	12.5	21.8	21.8
	NL2						
	Open-NL	0.0	0.0	6.6	18.4	32.7	18.9
A5	NAL-	0.0	0.0	6.6	20.7	27.1	26.6
	NL2						
	Open-NL	0.0	0.0	11.2	27.6	38.8	23.5
A6	NAL-	22.5	24.2	32.9	35.6	41.4	42.6
	NL2						
	Open-NL	23.6	31.4	37.8	39.1	43.2	34.0

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A7	NAL-	34.7	34.6	34.7	34.8	35.0	35.1
	NL2						
	Open-NL	37.5	37.5	37.5	37.5	37.5	37.5

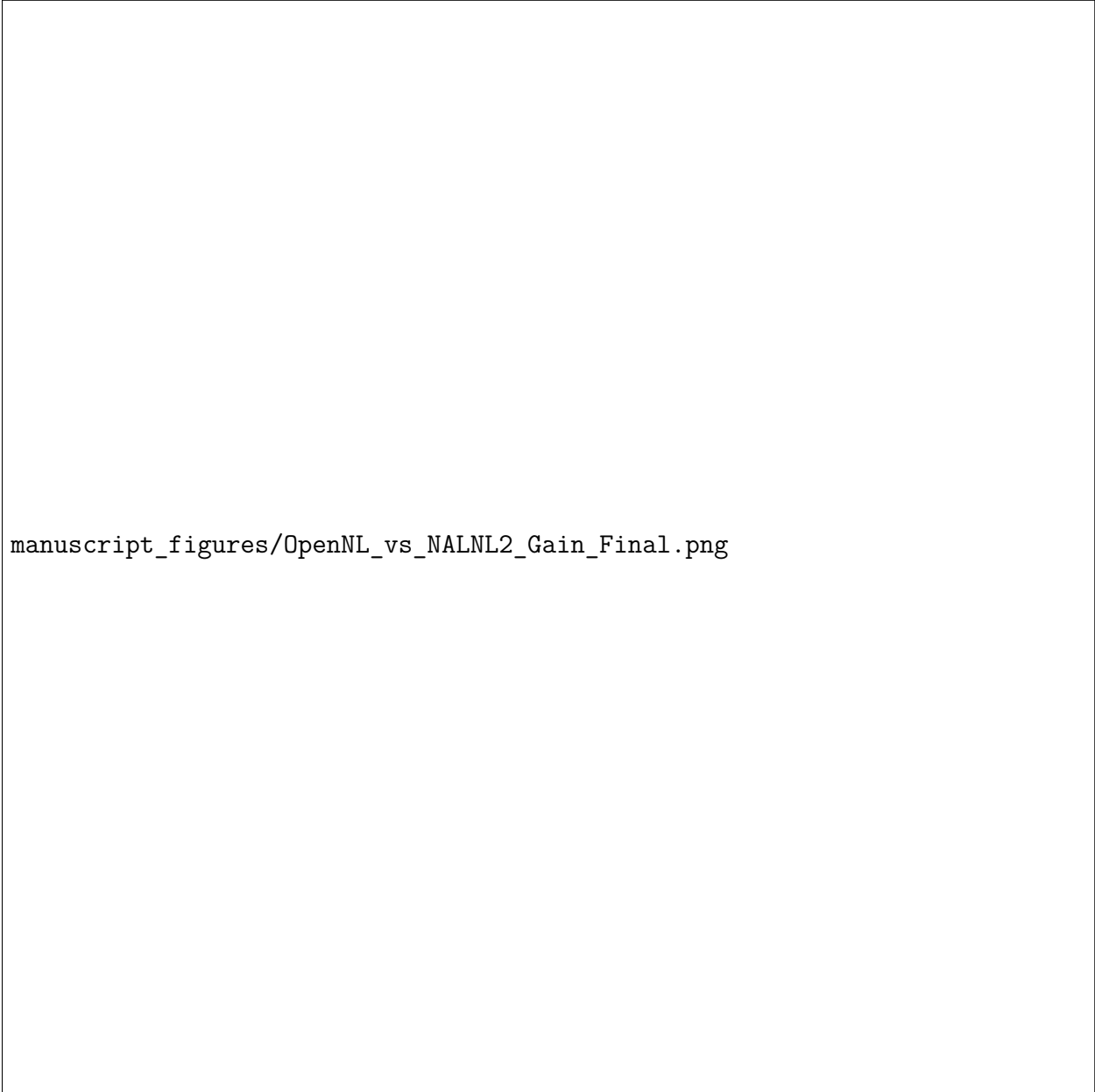
To prevent convergence bias, Open-NL’s C++ objective function avoids sparse-array Riemann approximations, dynamically interpolating the search array onto an internal 2048-point FFT frequency grid (0 to 22.05 kHz). For mild-to-moderate losses (A1, A3), Open-NL expands soft speech (50 dB inputs) slightly more aggressively than NAL-NL2 to maximize audibility within the safe physiological envelope, while compressing higher-level inputs to maintain loudness parity (**Figure 1**, **Figure 2**).

For extreme steeply sloping or profound losses (A4, A5), unmodified SII maximization drives substantial high-frequency gain. Although Open-NL integrates desensitization penalties to temper this drive, it still prescribes substantially more high-frequency gain than NAL-NL2 (e.g., +10.9 dB at 4 kHz for A4, and +11.7 dB for A5 at 65 dB SPL inputs; Table VII). In listeners with severe loss, reduced spectral resolution, elevated hearing thresholds, and cochlear dead regions account for comparable shares of speech recognition variance, with dead regions specifically blunting the benefit of restored high-frequency audibility (Ching, Dillon, & Byrne, 1998; Baer, Moore, & Kluk, 2002; Vestergaard, 2003; Souza et al., 2018; Moualed, Humphries, & Ramsden, 2018). While high prescribed gain increases physical audibility on paper, it severely degrades perceptual clarity if suprathreshold distortion is unmodeled (Margolis et al., 2025). However, enforcing blanket high-frequency suppression



manuscript_figures/OpenNL_vs_NALNL2_SII_Grouped.png

Figure 1: Comparison of ANSI SII (Raw Physical Audibility) vs Effective SII (Desensitized) for NAL-NL2 and Open-NL across 50, 65, and 80 dB SPL Inputs.



manuscript_figures/OpenNL_vs_NALNL2_Gain_Final.png

Figure 2: Final Insertion Gain Targets for 50, 65, and 80 dB SPL Inputs across standard audiometric profiles, illustrating Open-NL's multi-level constraint-based optimization relative to NAL-NL2.

based purely on pure-tone audiograms would penalize the majority of candidates who benefit from audibility (Cox et al., 2011, 2012; Pepler et al., 2015). Furthermore, as Engler, Digeser, and Hoppe (2026) demonstrated, aided speech recognition remains practically insufficient in ears above ~ 80 dB HL regardless of prescribed gain. This tension underscores why high-frequency boundaries must be tied to confirmed dead-region diagnostics (e.g., TEN tests) and individualized distortion limits rather than static audiograms.

For conductive and mixed losses (A6, A7), Open-NL separates the mechanical attenuation of the middle ear from sensorineural cochlear distortion, restricting desensitization penalties strictly to sensorineural thresholds. In profile A7 (pure conductive loss with a 50 dB air-bone gap), the optimizer prescribes an exact, flat 37.5 dB of linear gain across frequencies and levels via the 75% ABG restoration rule (yielding 1.00 SII at 3.85 sones). Crucially, this 75% restoration rule is an engineering choice adapted from clinical conventions (Johnson, 2013a; Scollie et al., 2005) to prevent hardware saturation, rather than an empirical preference optimum. This contrasts with well-supported heuristic targets like the 3.0:1 Compression Ratio bound (Stage 12), which is directly grounded in extensive empirical psychoacoustic data (Souza, 2002; Souza et al., 2006). However, as shown in the Supplementary Material (Table III), because Open-NL applies this bound as a soft penalty (P_{cr}), the raw drive to maximize SII in profound profiles (A4) completely overwhelms it, driving localized CRs as high as 12.00. This precisely illustrates why soft algorithmic penalties are insufficient to constrain pure intelligibility maximization, necessitating hard architectural bounds. For reverse-slope losses (A2), the SD-LFP constraint successfully limits low-frequency over-amplification, demonstrating how integrated constraints stabilize complex objective landscapes.

To model severe-loss distortion mathematically, Open-NL adapts the empirical desen-

463 sitization formulation of Johnson & Dillon (2011). Crucially, the engine isolates the pure
 464 sensorineural component ($T_{hl} = \max(0, T'_i - J_i)$) by subtracting the air-bone gap (J_i), and
 465 corrects a historical flaw in ANSI S3.5 implementations by restricting the internal cochlear
 466 noise floor calculation strictly to sensorineural loss ($X'_i = X_i + \max(0, T'_i - J_i)$), preventing
 467 conductive attenuation from falsely inflating internal noise. While the rigid clinical for-
 468 mula ($K'_{complete}$) introduces non-differentiable step boundaries that stall simplex optimizers,
 469 Open-NL’s optimizer evaluates intermediate solutions against a continuous mathematical
 470 relaxation:

$$K'_{smoothed} = K_i \cdot m, \quad \text{where } m = \frac{1}{1 + e^{0.075(T_{hl} - 66)}} \quad (1)$$

471 where K_i is raw audibility and m is the desensitization multiplier. This continuous relaxation
 472 permits smooth gradient descent, after which finalized targets are post-scored against the rigid
 473 piecewise $K'_{complete}$ formulation (detailed fully in Stage 7 and Stage 12 of the Supplementary
 474 Material).

475 C. Clinical Validation Protocols and Falsifiable Predictions

476 While synthetic evaluations verify mathematical behavior, translating Open-NL to clinical
 477 application mandates three non-negotiable validation hard gates to address the metric tautol-
 478 ogy and binaural summation boundaries: 1. **Real-Ear Measurement (REM) Verification:**
 479 Because physical ear-canal acoustics, leakage, and transducer roll-off decouple eardrum SPL
 480 from simulated targets (Dao et al., 2021), REM verification is mandatory. Empirical evi-
 481 dence robustly demonstrates that REM-verified fittings significantly outperform unverified
 482 first-fits on speech recognition and patient preference (Valente et al., 2018; Almufarrij, Dillon,

& Munro, 2021). **2. Individualized Broadband Loudness-Tolerance Safety Gates**

(trueLOUDNESS): Because monaural loudness models underestimate perceived binaural

broadband loudness by ~ 13 dB in 30–40% of impaired listeners (Denk et al., 2025; Moore et al.,

2014), clinical translation cannot rely on audiogram-derived monaural ceilings. Prior to any

behavioral testing, individualized binaural broadband loudness scaling (e.g., trueLOUDNESS

procedures; Oetting et al., 2016, 2017) or rigorous Uncomfortable Loudness Level (UCL)

verification must be administered to establish subject-specific safety caps and prevent acoustic

trauma. **3. Independent Waveform-Level Speech Recognition Benchmarks:** To

overcome the metric tautology of SII scoring, aided performance must be benchmarked using

independent speech-in-noise testing (e.g., matrix sentence tests or WIN/HINT) alongside

computational HASPI/HASQI modeling (Kates & Arehart, 2022). These evaluations must

reference conservative clinical controls (such as DSL v5.0 or NAL-NL2), which empirical

literature robustly favors in the 50–80 dB HL range (Mueller, 2005; Engler, Digeser, & Hoppe,

2026).

Beyond safety protocols, this framework yields a concrete, falsifiable clinical prediction:

because Open-NL’s uncalibrated A4 and A5 high-frequency targets exceed NAL-NL2 by

roughly 11 dB at 4000 Hz, they push far beyond historical comfort boundaries (Keidser et

al., 2012; Denk et al., 2025). We offer the following operational hypothesis: If adult listeners

with A4 or A5 audiometric profiles are fitted with real-ear verified Open-NL targets, $>80\%$

will exhibit immediate categorical loudness rejection—operationally defined as a rating of 6

(“Loud”) or 7 (“Uncomfortably Loud”) on the 7-point Categorical Loudness Scaling (CLS)

procedure (ISO 16832)—when presented with continuous broadband speech (e.g., ISTS) at

65 and 80 dB SPL, relative to a matched NAL-NL2 baseline. Empirically quantifying this

rejection threshold will provide the ground-truth data required to constrain distortion-aware
objective functions in future stochastic calibrations.

V. CONCLUSION

Open-NL provides a transparent, modular computational testbed for modeling, ablating,
and evaluating WDRC prescriptive heuristics natively within R. By coupling an explicitly
defined mathematical pipeline with an embedded C++ specific-loudness engine, the package
enables researchers to systematically inspect the trade-offs between audibility and physiological
loudness without relying on closed-source clinical software. As the framework evolves, it
provides the computational substrate needed to evaluate emerging multi-profile rationales such
as NAL-NL3 (Kitterick, Zakis, & Edwards, 2026) and to integrate individualized broadband
loudness summation metrics (Denk et al., 2025). As emphasized throughout, Open-NL is
strictly a research sandbox: its raw outputs must not be used for fitting hearing aids on
human patients, and any future translational research mandates Institutional Review Board
approval, mandatory real-ear verification, and individualized loudness-tolerance safety gates.

ACKNOWLEDGMENTS

The author wishes to thank the original developers of the R-project ecosystem and the
open-source contributors whose foundational work enabled the creation of this computational
toolkit.

AUTHOR DECLARATIONS

Conflict of Interest

The author declares no conflicts of interest.

Ethics Approval

The author declares that no animal subjects or human participants were involved in the development, theoretical simulation, or mathematical validation presented in this research. Because Open-NL is distributed strictly as a theoretical computational toolkit, any future application of the algorithm to human participants by independent investigators requires separate Institutional Review Board (IRB) approval, informed consent, and rigorous audiological safety protocols.

DATA AVAILABILITY

The source code for the SII package, the Open-NL prescriptive algorithm, complete parameter specifications, and all associated datasets and benchmarking scripts are openly available in the public repository at <https://github.com/r-gregmisc/SII> (v1.2.4; Git commit `b2b5ce0`; Archival DOI: 10.5281/zenodo.14963842; License: GPL-3.0). Standalone replication scripts generating all figures, tables, and sensitivity sweeps reported in this manuscript are located in the `reproducibility_scripts/` directory.

VI. REFERENCES

- Almufarrij, I., Dillon, H., & Munro, K. J. (2021). Does probe-tube verification of real-ear hearing aid amplification characteristics improve outcomes in adult hearing aid users? A systematic review and meta-analysis. *Trends in Hearing*, 25.
- Baer, T., Moore, B. C., & Kluk, K. (2002). Effects of low pass filtering on the intelligibility of speech in quiet for people with and without dead regions at high frequencies. *The Journal of the Acoustical Society of America*, 112(3), 1133-1144.
- Byrne, D., & Dillon, H. (1986). The National Acoustic Laboratories' (NAL) new procedure for selecting the gain and frequency response of a hearing aid. *Ear and Hearing*, 7(4), 257-265.
- Byrne, D., Parkinson, A., & Newall, P. (1990). Hearing aid gain and frequency response requirements for the severely/profoundly hearing impaired. *Ear and Hearing*, 11(1), 40-49.
- Byrne, D., Dillon, H., Ching, T. Y., Katsch, R., & Keidser, G. (2001). NAL-NL1 procedure for fitting nonlinear hearing aids: Characteristics and comparisons with other procedures. *Journal of the American Academy of Audiology*, 12(1), 37-51.
- Ching, T. Y., Dillon, H., & Byrne, D. (1998). Speech recognition of hearing-impaired listeners: Predictions from audibility and the limited role of high-frequency amplification. *The Journal of the Acoustical Society of America*, 103(2), 1128-1140.
- Ching, T. Y., Dillon, H., Katsch, R., & Byrne, D. (2001). Maximizing effective audibility in hearing aid fitting. *Ear and hearing*, 22(3), 212-224.
- Convery, E., & Keidser, G. (2011). Transitioning hearing aid users with severe and profound loss to a new gain/frequency response: Benefit, perception, and acceptance. *Journal of the American Academy of Audiology*, 22(3), 168-180.

Cox, R. M., Alexander, G. C., Johnson, J., & Rivera, I. (2011). Cochlear dead regions in typical hearing aid candidates: prevalence and implications for use of high-frequency speech cues. *Ear and Hearing*, 32(3), 339-348.

Cox, R. M., Johnson, J. A., & Alexander, G. C. (2012). Implications of high-frequency cochlear dead regions for fitting hearing aids to adults with mild to moderately severe hearing loss. *Ear and Hearing*, 33(5), 573-587.

Dao, A., Folkeard, P., Baker, S., Pumford, J., & Scollie, S. (2021). Fit-to-Targets and Aided Speech Intelligibility Index Values for Hearing Aids Fitted to the DSL V5-Adult Prescription. *Journal of the American Academy of Audiology*, 32(2), 90-98.

Denk, F., Oetting, D., Latzel, M., Bonsel, H., & Husstedt, H. (2025). Prevalence of excess binaural broadband loudness summation in the hearing-impaired population and implications for hearing aid gain targets. *PLOS ONE*, 20(3), e0319236. <https://doi.org/10.1371/journal.pone.0319236>

Engler, M., Digeser, F., & Hoppe, U. (2026). Speech recognition and real-ear-measured amplification in hearing-aid users with various grades of hearing loss. *International Journal of Audiology*, 65(7), 834–845. <https://doi.org/10.1080/14992027.2024.2426009>

Herzke, T., Kayser, H., Loshaj, F., Grimm, G., & Hohmann, V. (2017). OpenMHA—An open-source software platform for hearing aid research. *Trends in Hearing*, 21, 2331216517743250.

Hornsby, B. W., & Ricketts, T. A. (2006). The effects of hearing loss on the contribution of high- and low-frequency speech information to speech understanding. II. Sloping hearing loss. *The Journal of the Acoustical Society of America*, 119(3), 1752-1763.

Hornsby, B. W., Johnson, E. E., & Picou, E. M. (2011). Effects of degree and configuration

of hearing loss on the contribution of high- and low-frequency speech information to bilateral speech understanding. *Ear and Hearing*, 32(5), 543-555.

Humes, L. E., & Kidd, G. R. (2016). Speech recognition for multiple bands: Implications for the Speech Intelligibility Index. *The Journal of the Acoustical Society of America*, 140(3), 2019-2026.

Johnson, E. E. (2013a). Prescriptive Amplification Recommendations for Hearing Losses with a Conductive Component and Their Impact on the Required Maximum Power Output: An Update with Accompanying Clinical Explanation. *Journal of the American Academy of Audiology*, 24(6), 452-460.

Johnson, E. E. (2013b). An initial-fit comparison of two generic hearing aid prescriptive methods (NAL-NL2 and CAM2) to individuals having mild to moderately severe high-frequency hearing loss. *Journal of the American Academy of Audiology*.

Johnson, E. E., & Dillon, H. (2011). A comparison of gain for adults from generic hearing aid prescriptive methods: Impacts on predicted loudness, frequency bandwidth, and speech intelligibility. *Journal of the American Academy of Audiology*, 22(7), 441-459.

Kates, J. M., Arehart, K. H., Anderson, M. C., Kumar Muralimanohar, R., & Harvey, L. O. (2018). Using objective metrics to measure hearing aid performance. *Ear and Hearing*, 39(6), 1165-1175.

Kates, J. M., & Arehart, K. H. (2022). An overview of the HASPI and HASQI metrics for predicting speech intelligibility and speech quality for normal hearing, hearing loss, and hearing aids. *Hearing Research*, 424, 108593.

Kaur, M., Ramekers, D., & Knipper, M. (2023). Temporal bone pathology in reverse-slope audiograms: Reevaluating the structural basis of low-frequency hearing loss. *Hearing*

609 *Research*, 427, 108654.

610 Keidser, G., & Grant, F. (2001). Comparing loudness normalization (IHAF) with speech
611 intelligibility maximization (NAL-NL1) when implemented in a two-channel device. *Ear and*
612 *Hearing*.

613 Keidser, G., Dillon, H., Dyrland, O., Carter, L., & Hartley, D. (2007). Preferred low-
614 and high-frequency compression ratios among hearing aid users with moderately severe to
615 profound hearing loss. *Journal of the American Academy of Audiology*, 18(1), 17-33.

616 Keidser, G., Dillon, H., Carter, L., & O'Brien, A. (2012). NAL-NL2 empirical adjustments.
617 *Trends in Amplification*, 16(4), 211-223.

618 Keidser, G., Dillon, H., Flax, M., Ching, T., & Brewer, S. (2012). The NAL-NL2
619 prescription procedure. *Audiology Research*, 2(1), e24.

620 Kitterick, P. T., Zakis, J. A., & Edwards, B. (2026). Evolving the philosophy:
621 From the NAL rule to NAL-NL3. *International Journal of Audiology*, 65(6), 513-524.
622 <https://doi.org/10.1080/14992027.2026.42342663>

623 Lybarger, S. F. (1944). *US Patent No. 2,357,838*. Washington, DC: U.S. Patent and
624 Trademark Office.

625 Majdak, P., Hollmach, V., & Baumgartner, R. (2022). AMT: Auditory Modeling Toolbox.
626 *Acta Acustica*, 6, 19.

627 Malicka, A. N., Munro, K. J., Baer, T., Baker, R. J., & Moore, B. C. (2013). The effect
628 of low-pass filtering on identification of nonsense syllables in quiet by school-age children
629 with and without cochlear dead regions. *Ear and Hearing*, 34(3), 358-366.

630 Margolis, R. H., Hornsby, B. W. Y., Saly, G. L., & Wilson, R. H. (2025). Predicted and mea-
631 sured word-recognition scores unmask distortion in the impaired auditory system. *The Journal*

of the *Acoustical Society of America*, 157(2), 555–568. <https://doi.org/10.1121/10.0036461>

Moore, B. C., Glasberg, B. R., & Stone, M. A. (2010). Development of a new method for deriving initial fittings for hearing aids with multi-channel compression: CAMEQ2-HF. *International Journal of Audiology*, 49(3), 216-227.

Moore, B. C., Gibbs, A., Onions, G., & Glasberg, B. R. (2014). Measurement and modeling of binaural loudness summation for hearing-impaired listeners. *The Journal of the Acoustical Society of America*, 136(5), 2697-2708.

Moualed, D., Humphries, J., & Ramsden, J. D. (2018). Cochlear dead regions: Using the Threshold Equalising Noise (TEN) test to improve the assessment of potential cochlear implant candidates—The Oxford experience. *Clinical Otolaryngology*, 43(1), 384-387.

Mueller, H. G. (2005). Fitting hearing aids to adults using prescriptive methods: An evidence-based review of effectiveness. *Journal of the American Academy of Audiology*, 16(7), 448-460.

Oetting, D., Hohmann, V., Appell, J. E., Kollmeier, B., & Ewert, S. D. (2017). Restoring perceived loudness for listeners with hearing loss. *Ear and Hearing*, 38(1), 74-83.

Narayanan, S. K., Rye, P., Houmøller, S. S., et al. (2024). Difference in SII provided by initial fit and NAL-NL2 and its relation to self-reported hearing aid outcomes. *International Journal of Audiology*.

National Acoustic Laboratories. (2021). *NAL-NL2 software* [Computer software]. Sydney, Australia: National Acoustic Laboratories.

Pepler, A., Lewis, K., & Munro, K. J. (2015). Adult hearing-aid users with cochlear dead regions restricted to high frequencies: implications for amplification. *International Journal of Audiology*, 54(5), 297-306.

Pieper, I., Mauermann, M., Kollmeier, B., & Ewert, S. D. (2021). Toward an Individual Binaural Loudness Model for Hearing Aid Fitting and Development. *Frontiers in Psychology*, 12, 638662.

Plyler, P. N., & Fleck, E. L. (2006). The effects of high-frequency amplification on the objective and subjective performance of hearing instrument users with varying degrees of high-frequency hearing loss. *Journal of Speech, Language, and Hearing Research*, 49(3), 616-627.

Scollie, S., Seewald, R., Cornelisse, L., Moodie, S., Bagatto, M., Lurnagaray, D., Beaulac, S., & Pumford, J. (2005). The Desired Sensation Level multistage input/output algorithm. *Trends in Amplification*, 9(4), 159-197.

Souza, P. E. (2002). Effects of compression on speech acoustics, intelligibility, and sound quality. *Trends in Amplification*, 6(4), 131-165.

Souza, P. E., Jenstad, L. M., & Boike, K. T. (2006). Measuring the acoustic effects of compression amplification on speech in noise. *The Journal of the Acoustical Society of America*, 119(1), 41-44. <https://doi.org/10.1121/1.2108861>

Souza, P., Hoover, E., Blackburn, M., & Gallun, F. (2018). The characteristics of adults with severe hearing loss. *Journal of the American Academy of Audiology*, 29(8), 764-779.

Storey, L., Dillon, H., Yeend, I., & Wigney, D. (1998). The National Acoustic Laboratories' procedure for selecting the saturation sound pressure level of hearing aids: Experimental validation. *Ear and Hearing*, 19(4), 267-279.

Valente, M., Oeding, K., Brockmeyer, A., Smith, S., & Kallogjeri, D. (2018). Differences in word and phoneme recognition in quiet, sentence recognition in noise, and subjective outcomes between manufacturer first-fit and hearing aids programmed to NAL-NL2 using

678 real-ear measures. *Journal of the American Academy of Audiology*, 29(8), 706-721.

679 van Beurden, M., Boymans, M., van Geleuken, M., et al. (2021). Uni- And Bilateral
680 Spectral Loudness Summation and Binaural Loudness Summation With Loudness Matching
681 and Categorical Loudness Scaling. *International Journal of Audiology*, 60(2), 108-118.

682 Vestergaard, M. D. (2003). Dead regions in the cochlea: Implications for speech recognition
683 and applicability of articulation index theory. *International Journal of Audiology*, 42(5),
684 249-261.

685 Yoho, S. E., & Bosen, A. K. (2019). Individualized frequency importance functions for
686 listeners with sensorineural hearing loss. *The Journal of the Acoustical Society of America*,
687 146(2), 1184-1193.